an instant, relinquishing the energy borrowed in their creation. And the same is true for all of the other forms that energy and momentum can take—other particle eruptions and annihilations, wild electromagnetic-field oscillations, weak and strong force-field fluctuations—quantum-mechanical uncertainty tells us the universe is a teeming, chaotic, frenzied arena on microscopic scales. As Feynman once jested, "Created and annihilated, created and annihilated—what a waste of time."2 Since the borrowing and repaying on average cancel each other out, an empty region of space looks calm and placid when examined with all but microscopic precision. The uncertainty principle, however, reveals that macroscopic averaging obscures a wealth of microscopic activity.3 As we will see shortly, this frenzy is the obstacle to merging general relativity and quantum mechanics.

Quantum Field Theory

Over the course of the 1930s and 1940s theoretical physicists, led by the likes of Paul Dirac, Wolfgang Pauli, Julian Schwinger, Freeman Dyson, Sin-Itiro Tomonaga, and Feynman, to name a few, struggled relentlessly to find a mathematical formalism capable of dealing with this microscopic obstreperousness. They found that Schrödinger's quantum wave equation (mentioned in Chapter 4) was actually only an approximate description of microscopic physics—an approximation that works extremely well when one does not probe too deeply into the microscopic frenzy (either experimentally or theoretically), but that certainly fails if one does.

The central piece of physics that Schrödinger ignored in his formulation of quantum mechanics is special relativity. In fact, Schrödinger did try to incorporate special relativity initially, but the quantum equation to which this led him made predictions that proved to be at odds with experimental measurements of hydrogen. This inspired Schrödinger to adopt the time-honored tradition in physics of divide and conquer: Rather than trying, through one leap, to incorporate all we know about the physical universe in developing a new theory, it is often far more profitable to take many small steps that sequentially include the newest discoveries from the forefront of research. Schrödinger sought and found a mathematical framework encompassing the experimentally discovered wave-particle duality, but